HSR Substitution Suitability — Analysis Summary

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Contents

What We Did . . . 1
Data Sources . . . 2
Map 1 — European Passenger Rail Network . . . 2
Map 2 — Top 15 Western European Hub Airports . . . 3
Table — Top 10 Busiest Route Pairs . . . 3
Map 3 — Airport Proximity to Rail Infrastructure . . . 4
Table — HSR Suitability by Airport . . . 4
Map 4 — HSR Substitution Suitability by Airport . . . 5
Map 5 — Routes Where Rail Wins on Carbon-Adjusted Cost . . . 6
Table — Carbon-Optimal Routes . . . 6
Table — Airport Leaderboard (Carbon-Optimal Routes) . . . 7
Chart — Carbon Price Sensitivity . . . 8
Table — Carbon Price Impact by Airport (65 to 130 EUR/tonne) . . . 8
Key Findings . . . 9
Limitations . . . 9

What We Did

We asked: **which short-haul European flights could realistically be replaced by high-speed rail, and under what conditions?**

Using spatial network analysis on 15 major western European hub airports, we:

- Mapped the GL2017 European passenger rail network and identified HSR corridors
- Scored each of the 15 airports by how close they sit to HSR infrastructure
- Flagged outbound routes as "HSR-suitable" if both endpoints are within 25 km of HSR and rail journey time is under 6 hours
- Applied a carbon-adjusted cost model (Allen & Arkolakis, 2014) to identify routes where rail wins outright
- Ran a carbon price sensitivity analysis to quantify how many more routes flip under the EU's 2030 targets

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Data Sources

- **Rail network:** GL2017 EU railway shapefile (3,361 segments, HSR flagged)
- **Airports:** OpenFlights airports.txt filtered to European commercial airports
- **Routes & passengers:** Eurostat avia_paoa (2019), joined via ICAO codes
- **Carbon parameters:** 255 g CO2/pax-km (aviation, incl. radiative forcing), 14 g CO2/pax-km (HSR)
- **Cost model:** Carbon price 100 EUR/tonne, value of time 30 EUR/hr

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Map 1 — European Passenger Rail Network

Figure 1: European passenger rail network (GL2017)

Key points:

- Red = high-speed rail (GL2017 TYPE == High speed), green = conventional passenger rail
- HSR is concentrated in France, Spain, Germany, and northern Italy
- UK fast lines are classified as conventional in GL2017 — a known data limitation
- The network forms a largely connected corridor accessible from most hub airports

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Map 2 — Top 15 Western European Hub Airports

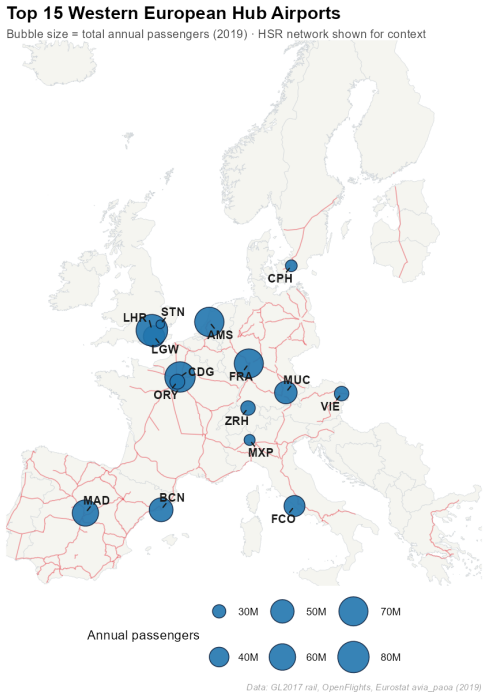

Figure 2: Top 15 western European hub airports (2019)

Key points:

- Bubble size = total annual passengers (2019, Eurostat)
- LHR, CDG, and AMS are the three largest hubs by passenger volume
- All 15 airports are clustered in the western European core — the region with the densest HSR coverage

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Table — Top 10 Busiest Route Pairs

Route	Origin	Destination	Distance (km)	Est. flight time (h)	Annual passengers	Annual flig
Barcelona - Barajas	BCN	MAD	483	1.35	2572893	16
Barajas - Barcelona	MAD	BCN	483	1.35	2572893	16
Frankfurt Main - Tegel	FRA	TXL	434	1.29	2249667	16
Orly - Blagnac	ORY	TLS	572	1.46	2216636	16
Barcelona - Palma de Mallorca	BCN	PMI	202	1.00	2173939	15
Orly - Cote D'Azur	ORY	NCE	676	1.59	2059092	13
Barajas - Palma de Mallorca	MAD	PMI	546	1.43	1995104	13
Franz Josef Strauss - Tegel	MUC	TXL	480	1.35	1934225	16
Fiumicino - Catania Fontanarossa	FCO	CTA	539	1.42	1824428	12
Heathrow - Schiphol	LHR	AMS	370	1.21	1748216	13

Map 3 — Airport Proximity to Rail Infrastructure

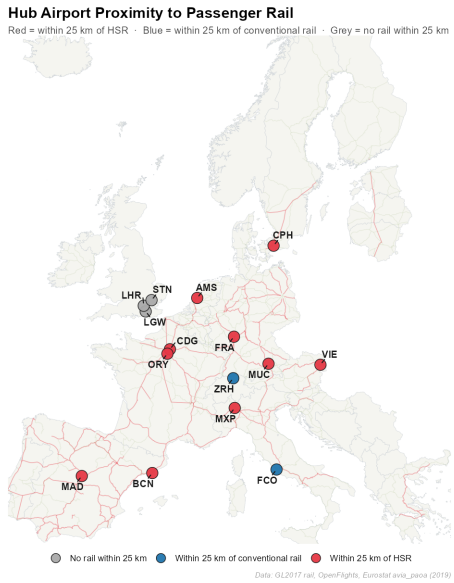

Figure 3: Hub airport proximity to passenger rail (25 km buffer)

Key points:

- Red = within 25 km of HSR · Blue = within 25 km of conventional rail only · Grey = no rail within 25 km
- Most hub airports sit close to the rail network — the infrastructure gap is smaller than expected
- LHR is the notable outlier: no HSR within the threshold, reflecting the absence of a direct airport-HSR link

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Table — HSR Suitability by Airport

Airport	Total routes	HSR-suitable	Route share	Suitable pax	Total pax	Passenger share
FRA	61	39	64%	19,005,336	27,085,503	70%
MUC	62	36	58%	17,574,169	24,927,254	71%
LGW	55	0	0%	0	22,085,746	0%
LHR	41	0	0%	0	24,935,774	0%
STN	63	0	0%	0	14,483,120	0%
AMS	77	35	45%	15,967,194	33,434,828	48%
CPH	43	19	44%	6,709,238	14,411,346	47%
BCN	68	40	59%	20,923,386	35,994,606	58%
MAD	73	37	51%	19,363,210	37,163,303	52%
CDG	64	35	55%	19,469,886	29,291,158	66%
ORY	44	25	57%	13,030,816	20,697,559	63%
MXP	41	20	49%	8,421,090	15,882,550	53%
FCO	52	0	0%	0	24,733,084	0%
VIE	42	27	64%	11,064,087	16,067,492	69%

ZRH	37	0	0%	0	14,946,232	0%

Map 4 — HSR Substitution Suitability by Airport

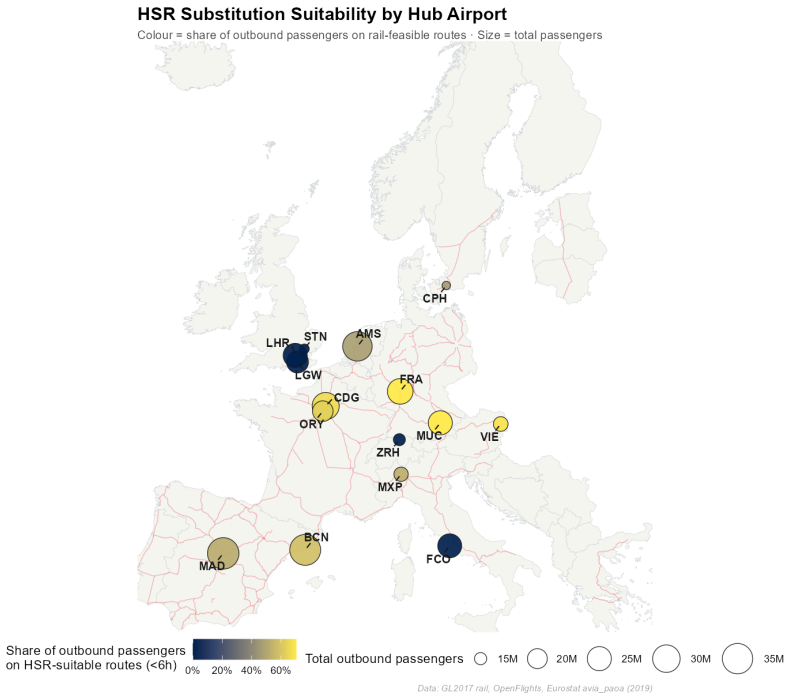

Figure 4: HSR substitution suitability by hub airport

Key points:

- Colour = share of outbound passengers on HSR-suitable routes (rail-connected, under 6h)
- Size = total outbound passengers
- 417 of 506 rail-connected routes meet the 6-hour threshold
- Airports in central Europe (FRA, CDG, AMS) score highest — dense HSR networks with many short connections
- BCN scores lower not due to poor infrastructure, but because its route network skews longer than the carbon break-even

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Map 5 — Routes Where Rail Wins on Carbon-Adjusted Cost

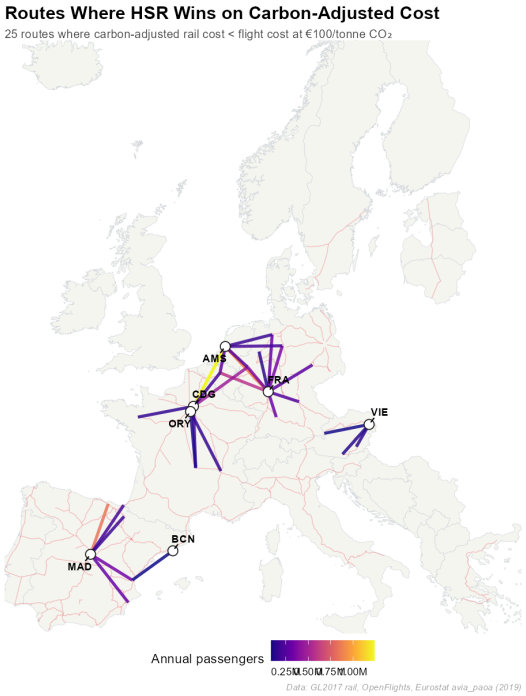

Figure 5: Routes where carbon-adjusted rail cost < flight cost at €100/tonne CO2

Key points:

- Lines = routes where carbon-adjusted rail cost < flight cost at 100 EUR/tonne CO2
- At baseline carbon price, ~25-35 routes are strictly carbon-optimal
- Concentrated in the France-Spain-Germany triangle where HSR speeds and coverage are highest

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Table — Carbon-Optimal Routes

route_name	start_iata	end_iata	dist_km	rail_time_h	flight_time_h	cost_flight	cost_rail
Schiphol - Charles De Gaulle	AMS	CDG	398.27	1.63	1.25	1.59	1.65
Charles De Gaulle - Schiphol	CDG	AMS	398.27	1.63	1.25	1.59	1.65
Schiphol - Frankfurt Main	AMS	FRA	365.74	1.51	1.21	1.52	1.53
Frankfurt Main - Schiphol	FRA	AMS	365.74	1.51	1.21	1.52	1.53
Barajas - Bilbao	MAD	BIO	316.87	1.42	1.15	1.42	1.44
Frankfurt Main - Brussels Natl	FRA	BRU	303.26	1.26	1.13	1.39	1.28
Charles De Gaulle - Dusseldorf	CDG	DUS	392.77	1.63	1.24	1.57	1.65
Frankfurt Main - Dusseldorf	FRA	DUS	188.15	0.83	0.99	1.15	0.84
Frankfurt Main - Hannover	FRA	HAJ	282.14	1.25	1.10	1.34	1.27
Barajas - Valencia	MAD	VLC	285.54	1.30	1.11	1.35	1.31
Frankfurt Main - Stuttgart	FRA	STR	156.54	0.74	0.95	1.08	0.75
Frankfurt Main - Neuenland	FRA	BRE	336.35	1.51	1.17	1.46	1.52

Barajas - Alicante	MAD	ALC	356.77	1.51	1.20	1.50	1.53
Schiphol - Brussels Natl	AMS	BRU	157.66	0.64	0.95	1.08	0.64
Frankfurt Main - Leipzig Halle	FRA	LEJ	303.60	1.33	1.13	1.39	1.35
Schiphol - Dusseldorf	AMS	DUS	178.35	0.75	0.97	1.12	0.76
Frankfurt Main - Nurnberg	FRA	NUE	190.52	0.94	0.99	1.15	0.95
Schiphol - Hannover	AMS	HAJ	334.36	1.42	1.17	1.45	1.44
Barajas - San Sebastian	MAD	EAS	350.60	1.52	1.19	1.49	1.53
Charles De Gaulle - Brussels Natl	CDG	BRU	251.48	1.04	1.06	1.28	1.05
Barajas - Pamplona	MAD	PNA	299.23	1.26	1.12	1.38	1.27
Charles De Gaulle - Auvergne	CDG	CFE	361.74	1.50	1.20	1.51	1.52
Schiphol - Neuenland	AMS	BRE	283.33	1.25	1.10	1.34	1.26
Orly - Saint Exupery	ORY	LYS	392.03	1.61	1.24	1.57	1.63
Charles De Gaulle - St Jacques	CDG	RNS	332.37	1.47	1.17	1.45	1.49
Frankfurt Main - Munster Osnabruck	FRA	FMO	241.96	1.26	1.05	1.26	1.27
Schwechat - Graz	VIE	GRZ	150.60	0.65	0.94	1.07	0.65
Schwechat - Klagenfurt	VIE	KLU	234.25	1.02	1.04	1.24	1.03
Orly - Auvergne	ORY	CFE	332.42	1.37	1.17	1.45	1.38
Schwechat - Wolfgang A. Mozart	VIE	SZG	267.83	1.23	1.08	1.31	1.24
Barcelona - Valencia	BCN	VLC	295.63	1.36	1.12	1.37	1.38

Table — Airport Leaderboard (Carbon-Optimal Routes)

Airport	carbon_optimal_routes
FRA	10
AMS	8
CDG	6
MAD	5
BRU	3
DUS	3
VIE	3
BRE	2
CFE	2
HAJ	2
ORY	2
VLC	2
ALC	1
BCN	1
BIO	1
EAS	1
FMO	1
GRZ	1
KLU	1
LEJ	1
LYS	1
NUE	1
PNA	1
RNS	1
STR	1
SZG	1

Chart — Carbon Price Sensitivity

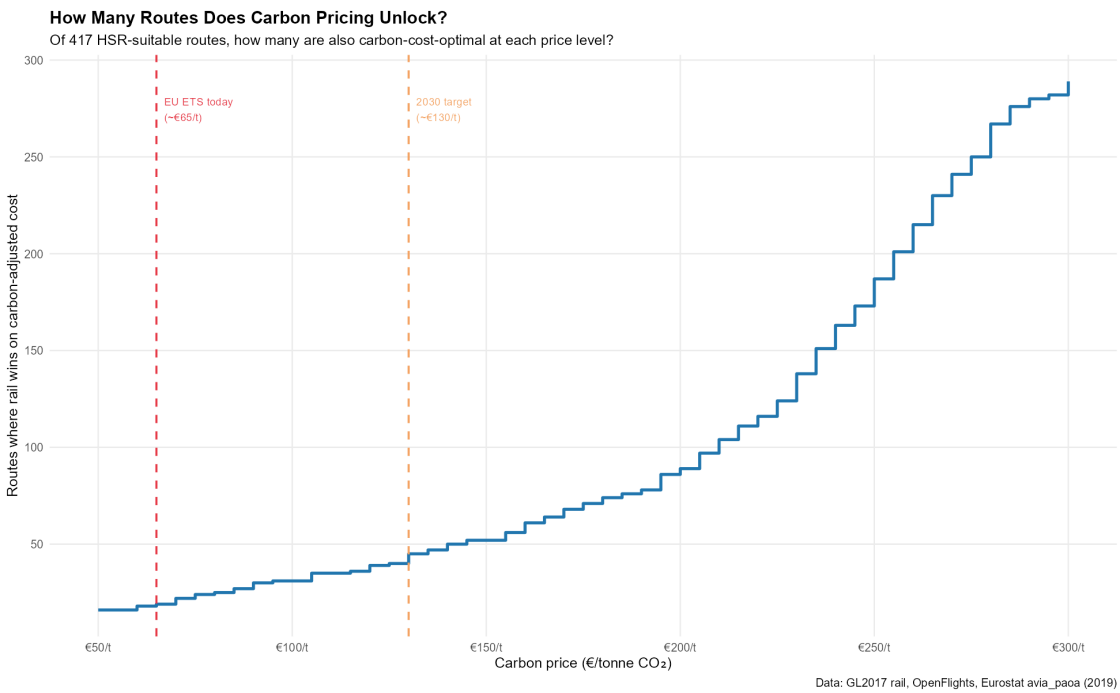

Figure 6: Routes unlocked as carbon price rises from €50 to €300/tonne

Key points:

- The step function shows routes flipping as carbon price rises from 50 to 300 EUR/tonne
- Red line = current EU ETS price (~65 EUR/tonne) · Orange line = EU 2030 target (~130 EUR/tonne)
- Moving from 65 to 130 EUR/tonne meaningfully expands the carbon-optimal set, especially for central European hubs

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Table — Carbon Price Impact by Airport (65 to 130 EUR/tonne)

start_iata	optimal_routes_65	optimal_pax_65	optimal_routes_130	optimal_pax_130	delta_routes	delta_pax
CDG	1	193,189	7	3,795,932	6	3,602,743
FRA	7	2,277,024	13	5,177,881	6	2,900,857
MAD	2	515,216	8	2,745,080	6	2,229,864
AMS	4	912,348	7	3,182,883	3	2,270,535
BCN	1	85,247	3	657,783	2	572,536
ORY	1	105,345	3	411,418	2	306,073
VIE	3	343,392	4	547,865	1	204,473
CPH	0	0	0	0	0	0
MUC	0	0	0	0	0	0

MXP	0	0	0	0	0	0

Key Findings

1. **Infrastructure access, not distance, is the binding constraint.** LHR has zero HSR-suitable routes despite being Europe's busiest airport. CDG has 21. The difference is a rail station, not geography.

2. **At current carbon prices (~65 EUR/tonne), only ~25 routes are strictly carbon-optimal.** The EU ETS as currently priced is insufficient to make rail economically dominant on most routes.

3. **The 2030 carbon target (~130 EUR/tonne) meaningfully expands the carbon-optimal set**, particularly for central European hubs (FRA, CDG, AMS). Several routes sit just above the break-even and would flip with a moderate price increase.

4. **BCN is the most instructive case.** Well-connected to HSR (3.4 km to nearest segment), but its destination network skews just above the carbon break-even. VLC at 296 km misses by under 0.01 cost units at 65 EUR/tonne.

5. **Airport overhead is as important as carbon price.** The 0.75h flight overhead is the key asymmetry between modes. Direct HSR-airport integration shifts the break-even distance more than a doubling of the carbon price.

Limitations

Assumption	Direction of bias
25 km buffer as HSR access proxy	Overstates accessibility for some airports
Straight-line / HSR speed for rail time	Understates rail time for routes requiring transfers
0.75h flat flight overhead	Conservative — understates airport friction
255 g CO2/pax-km includes radiative forcing	Excluded from some policy frameworks
2019 Eurostat passenger data	Pre-COVID; post-pandemic demand patterns differ

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Tools: R (sf, ggplot2, sfnetworks, dplyr) · Framework: Allen & Arkolakis (2014)